Lab Report 09

Course title: Computer Network Laboratory
Course code: CSE-3202
Session: July - December 2026

Date of Submission: *21 October, 2026*



Submitted to-

Md. Mahabubur Rahman
Lecturer
Department of Computer Science and Engineering
Bangladesh University of Professionals

Tanjim Taharat Aurpa
Assistant Professor
Department of Data Science and Engineering
University of Frontier Technology, Bangladesh

ID	Name
24524203082	MAHMUDA

Title: Implementation of Standard Access Control List (ACL)

Objective:

- To understand the concept and working principle of ACL.
- To configure a standard ACL on a router.
- To block traffic from a specific PC using its IP address.
- To test network connectivity before and after applying the ACL.
- To remove the ACL configuration and restore normal communication.

Description:

One router, two switches and four PCs were added to Cisco Packet Tracer. Two PCs were connected to each switch and each switch was connected to the router.

IP addresses, subnet masks and default gateways were configured for all PCs. The first LAN used the network 192.168.10.0, with gateway 192.168.10.1. The second LAN used the network 192.168.20.0 with gateway 192.168.20.1.

A standard ACL was created on the router to block traffic from the PC with IP address 192.168.10.10 while allowing traffic from other sources.

The ACL was applied inbound on interface GigabitEthernet0/0 using the ip access-group command. This allowed the router to filter incoming packets based on their source IP addresses.

Connectivity was tested using the ping command. Traffic from the blocked PC was denied, while traffic from other permitted sources was allowed.

Finally, the ACL was removed from the interface using the no ip access-group command, and the ACL itself was deleted using the no access-list 10 command. Connectivity was tested again.

Connection:

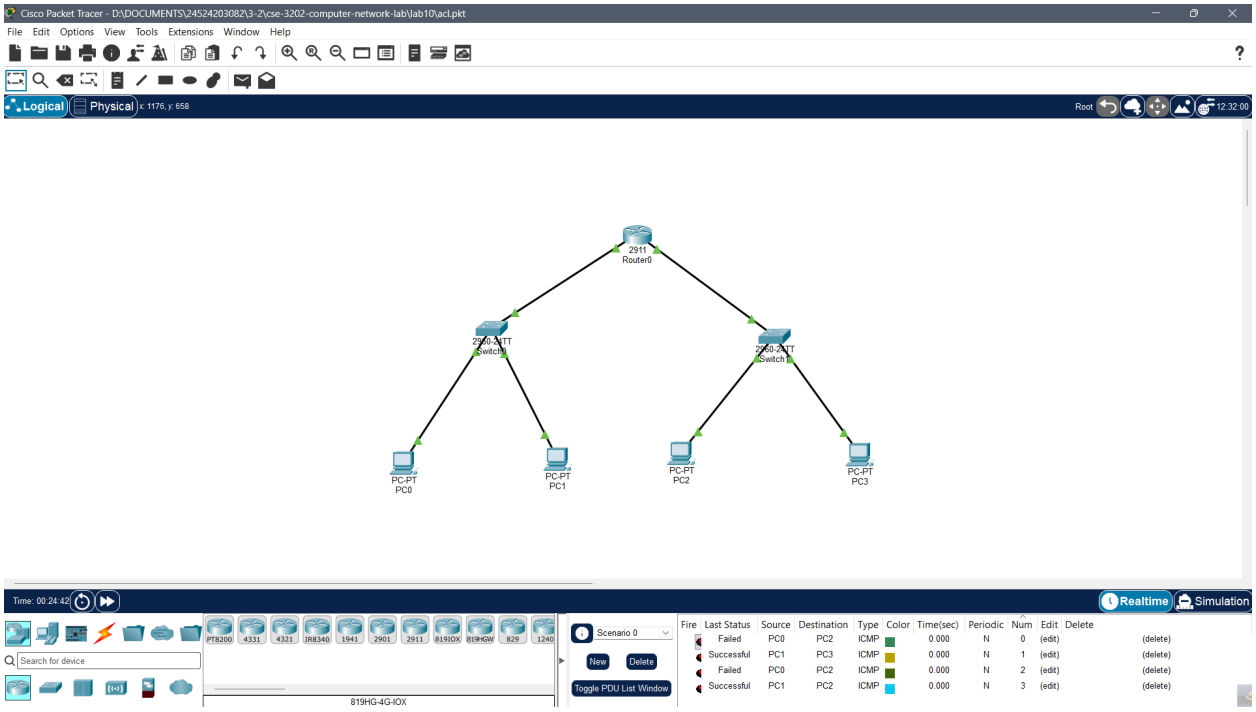


Figure 1: Network Topology for Standard ACL

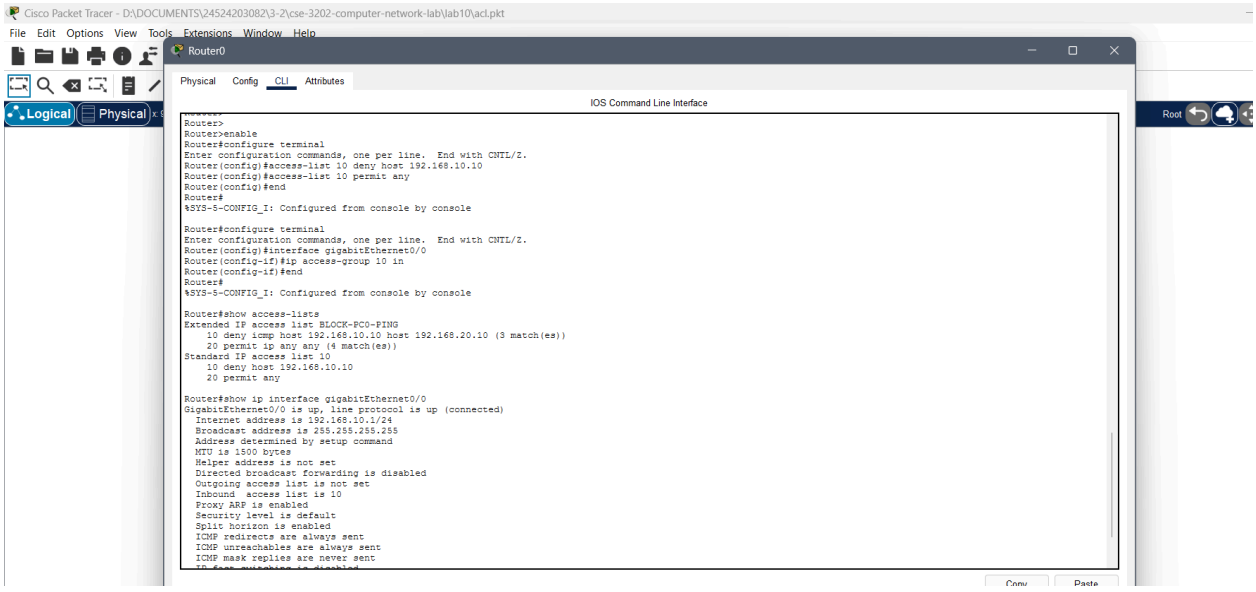
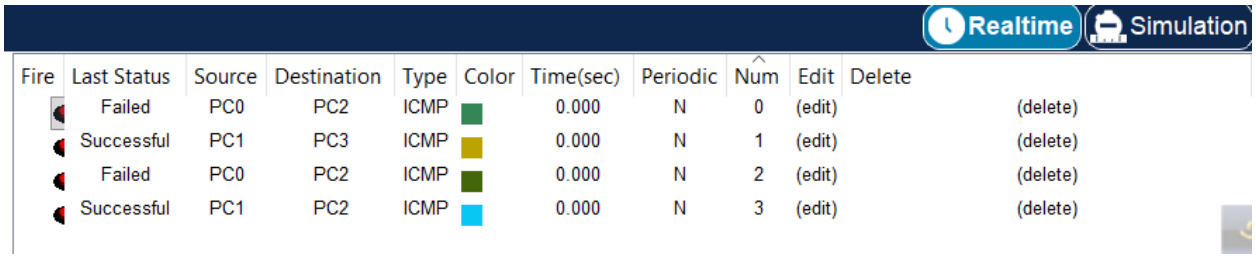


Figure 2: ACL Configuration on the Router



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC3	ICMP		0.000	N	1	(edit)	(delete)
	Failed	PC0	PC2	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	3	(edit)	(delete)

Figure 3: Connectivity Test After Applying ACL

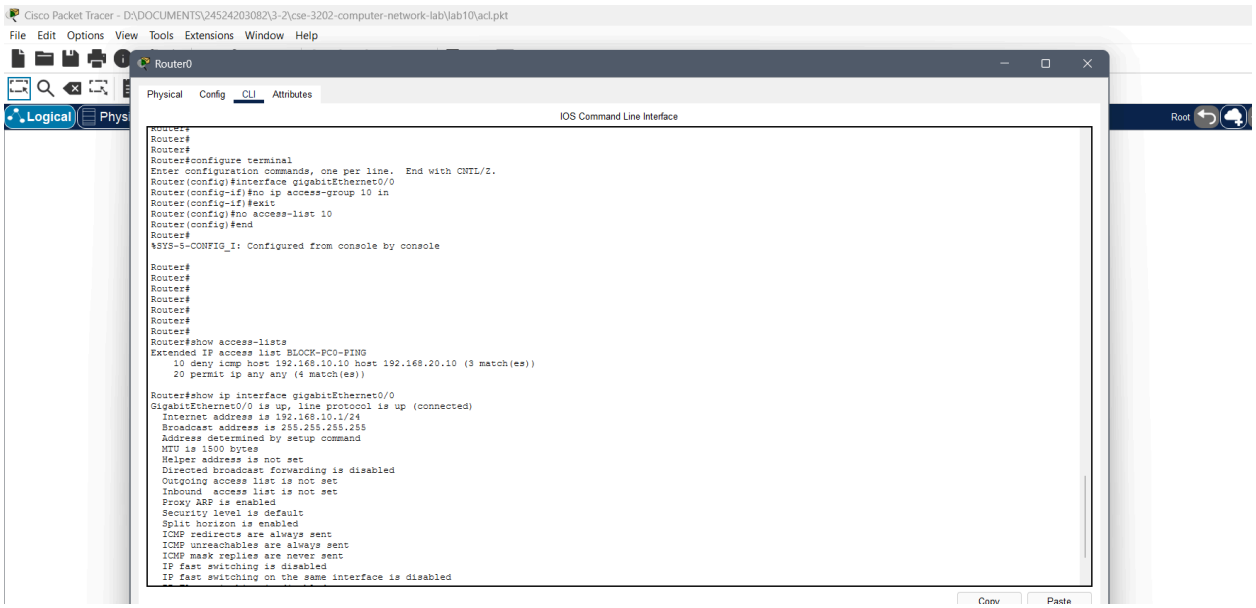
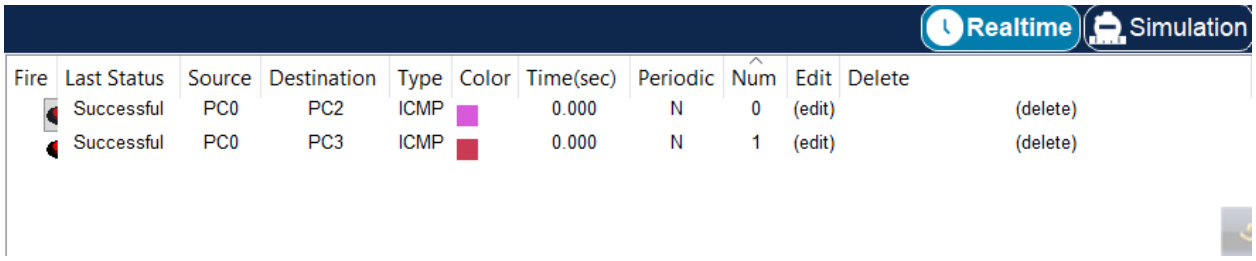


Figure 4: Removing the ACL Configuration



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC3	ICMP		0.000	N	1	(edit)	(delete)

Figure 5: Connectivity Test After Removing ACL